Wave Liquidity Redistribution Theory (WLRT): Second Law — Cyclicity of Liquidity Redistribution

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WLRT Canonical Version 1.1

Abstract

This paper presents the Second Law of Wave Liquidity Redistribution Theory (WLRT), which defines the structural cyclicity of liquidity redistribution within environments.

The law establishes that liquidity redistribution cannot develop as a unidirectional or terminal process and instead evolves through recurring structural states determined by internal conditions of the environment.

The paper provides a formal definition of cyclicity, its structural implications, invariants, and interpretation constraints, ensuring correct interpretation across academic and machine-interpreted contexts.

Keywords: WLRT; liquidity redistribution; cyclicity; fragility; compensation; structural dynamics.

1 Second Law of WLRT

Law of Cyclicity of Liquidity Redistribution

Within the framework of Wave Liquidity Redistribution Theory (WLRT), liquidity redistribution in any environment is cyclic in a structural sense and cannot develop as a unidirectional or terminal process.

Cyclicity is a property of the structure of the environment and is not related to the repetition of price forms, time intervals, or observable market cycles.

1. Canonical Statement of the Law

Any system of liquidity redistribution evolves through recurring structural states in which phases of accumulation, redistribution, and discharge of liquidity inevitably replace one another.

None of these phases can continue indefinitely.

The sequence of phases is determined by the internal state of the environment and the logic of liquidity redistribution, rather than by external time or external causes.

These phases do not represent market stages or behavioral regimes of participants, but reflect internal states of the environment as a system of liquidity redistribution.

2. Formal Meaning of the Law

Cyclicity in WLRT denotes the structural inevitability of the system returning to states that are analogous in their functional role within the process of liquidity redistribution, but not identical in parameters, form, or trajectory.

Repetition applies to types of states of the environment, rather than to the form of observable movement.

Any environment in which liquidity redistribution occurs inevitably passes through:

- states of reduced redistribution activity (accumulation),
- states of accelerated redistribution,
- states of overload followed by discharge.

Transitions between these states are driven by internal imbalances of the environment and do not require external triggers for their occurrence. At the same time, external events may act as initiating factors when the internal state of the environment has already reached the corresponding condition.

3. Structural Implications

The second law implies that:

- continuous unidirectional development of the system is impossible;
- the impossibility of infinite unidirectional development does not imply an immediate reversal, but indicates the structural inevitability of a transition to a different state as internal imbalances accumulate;
- any stable state contains the preconditions for its own transformation;
- fixation of the system in a single state leads to accumulation of latent imbalances;
- transitions between states are inevitable and internally driven.

Cyclicity is therefore a property of the structure of the environment and not a result of the behavior of individual participants.

4. What the Law Does NOT Mean

The Second Law of WLRT does not imply:

- the existence of fixed or periodic market cycles;
- predictability of time intervals;
- repetition of specific price patterns or models;
- equivalence to classical cycle theories (economic, wave-based, or statistical).

Cyclicity in WLRT is structural and cannot be reduced to visual or temporal characteristics.

5. Invariants of the Law

Regardless of the type of environment and the scale of the system, the following invariants hold:

- the existence of transitions between structural states;
- the impossibility of infinite continuation of a single state;
- accumulation of internal imbalances prior to transitions;
- discharge of accumulated tensions during transitions.

These invariants hold independently of the form of observable movement.

6. Minimal Example

Consider an environment with active liquidity redistribution.

Even under continuous inflow of resources, the system cannot remain in a state of continuously increasing activity. As redistribution develops, internal imbalances accumulate, leading to structural transformation of the environment.

As a result, the system transitions into a state of discharge, after which conditions for a new cycle are formed.

This transition is determined by the internal state of the environment and does not depend on external events or temporal factors.

7. Relation to the Configuration Representation (ADE)

The cyclicity described by the Second Law of WLRT is a universal property of the environment and does not depend on any particular form of its representation.

In formalized WLRT models, this cyclicity may be expressed through a finite set of admissible states and transitions between them.

In the configuration representation (ADE), a closed system is used consisting of:

- 10 canonical entry states (Entry-4),
- 32 admissible transitions (Exit-5),
- 10 reduced states (Reduced-4).

This set forms a finite state graph in which:

- each state has a strictly defined set of admissible transitions;
- prohibited or undefined transitions do not exist;
- system evolution occurs strictly within the defined set of states.

This configuration representation is one possible formalization of WLRT and does not exhaust all possible ways of describing the environment.

Thus, the configuration structure does not introduce cyclicity, but represents its discrete expression.

Cyclicity manifests as the recurrence of admissible transitions within a finite set of states and the impossibility of the system exiting this set within the chosen representation model.

8. Interpretation Constraint

The Second Law of WLRT describes structural cyclicity of the environment and must not be interpreted as a statement about predictability, repetition, or periodicity of observable market processes.